CP(M) 2026 Invigilated MiniZinc Assessment, mock paper: sample answers at about band B

This document gives a set of answers to the mock paper at about band B, written as a student might write them in the assessment. The answers are substantially correct but contain deliberate gaps, which the comment at the end identifies and relates to the band criteria. Line numbers refer to the appendices of the paper.

Question 1.1. (D) on line 55 makes nothing burning at turn 0, which is the starting state. Then at each turn, (E) makes `spread[t, v]` true if v was burning at turn $t-1$ or has a neighbour that was, which is the spreading step. (F) makes `burnt[t, v]` true if v is in `spread[t, v]` or was chosen at turn t. So if v is burning at the end of turn t, then `burnt[t, v]` is true, and by induction this holds for every turn t.

Question 1.2. No. On the Petersen graph, which has diameter 2, the fire from the vertex chosen at turn 1 reaches every vertex by the spread at turn 3. So at turn 3 every vertex has `spread[3, v]` true, and the modified (H) forbids any choice. But (A)-(C) need a vertex to be chosen at turn 3, so there is no solution with `turns` $= 3$, and the model is unsatisfiable even though $b = 3$.

Question 2.1. `y[0]`, `y[1]`, `y[2]` and `y[3]` are 1, so `turns` $= 4$ by (C). `burn_turn` is 1 at vertex 4, 2 at vertex 10, 3 at vertex 14 and 4 at vertex 16, and 0 elsewhere. The output is `sequence = [4, 10, 14, 16]`. For $u = 7$, $d(7, 4) = 3 \leq 3$, so `x[4, 3]` makes the sum in (E) at least 1. For $u = 13$, $d(13, 14) = 1 \leq 1$, so `x[14, 1]` does.

Question 2.2. The expression multiplies `turns` by `x[v, r]`, and both are variables, so it is not linear and cannot be used in an ILP. $M =$ `UB` is large enough because `UB` is the largest value that `turns` can take.

Question 3, first part. In Appendix A, replace (A) on line 44 by `constraint forall(t in 1..UB)(count(v in 1..n)(burn_turn[v] = t) <= k);` and delete (C) on line 50, since it no longer holds. In Appendix B, change (A) on line 53 to `constraint forall(r in RADIUS)(sum(v in 1..n)(x[v, r]) <= k * y[r]);`. Add `int: k;` to both models.

Question 3, second part. Choose 3 and 8 at turn 1, then 12 and 15 at turn 2, and nothing at turn 3. Then 1-5 and 6-10 are burning from the first two fires and 11-13 and 14-16 from the second two. Two turns are not enough because they can only burn 8 vertices.

Comment. All the answers are substantially correct, but several justifications have gaps.

- **Question 1.1** argues only one direction: it shows that `burnt[t, v]` is true for burning vertices, but not that it is false for the others, which is why (E) and (F) are equivalences. The induction is also only named, not set out.
- **Question 1.2** gives a correct counterexample. It should explain why `turns` must be 3, namely that $b = 3$ and `UB` $=$ rad $+ 1 = 3$, and it doesn't say what the deleted disjunct was for.
- **Question 2.2** states the reason for $M =$ `UB` without saying which inequality needs it, or why.
- **Question 3:** the reason for deleting (C) ("it no longer holds") is too brief, and the count of

8 vertices is asserted without explanation.

These are the "gaps in at most a few justifications" of the B band.